

COMP 3721

Introduction to Data Communications

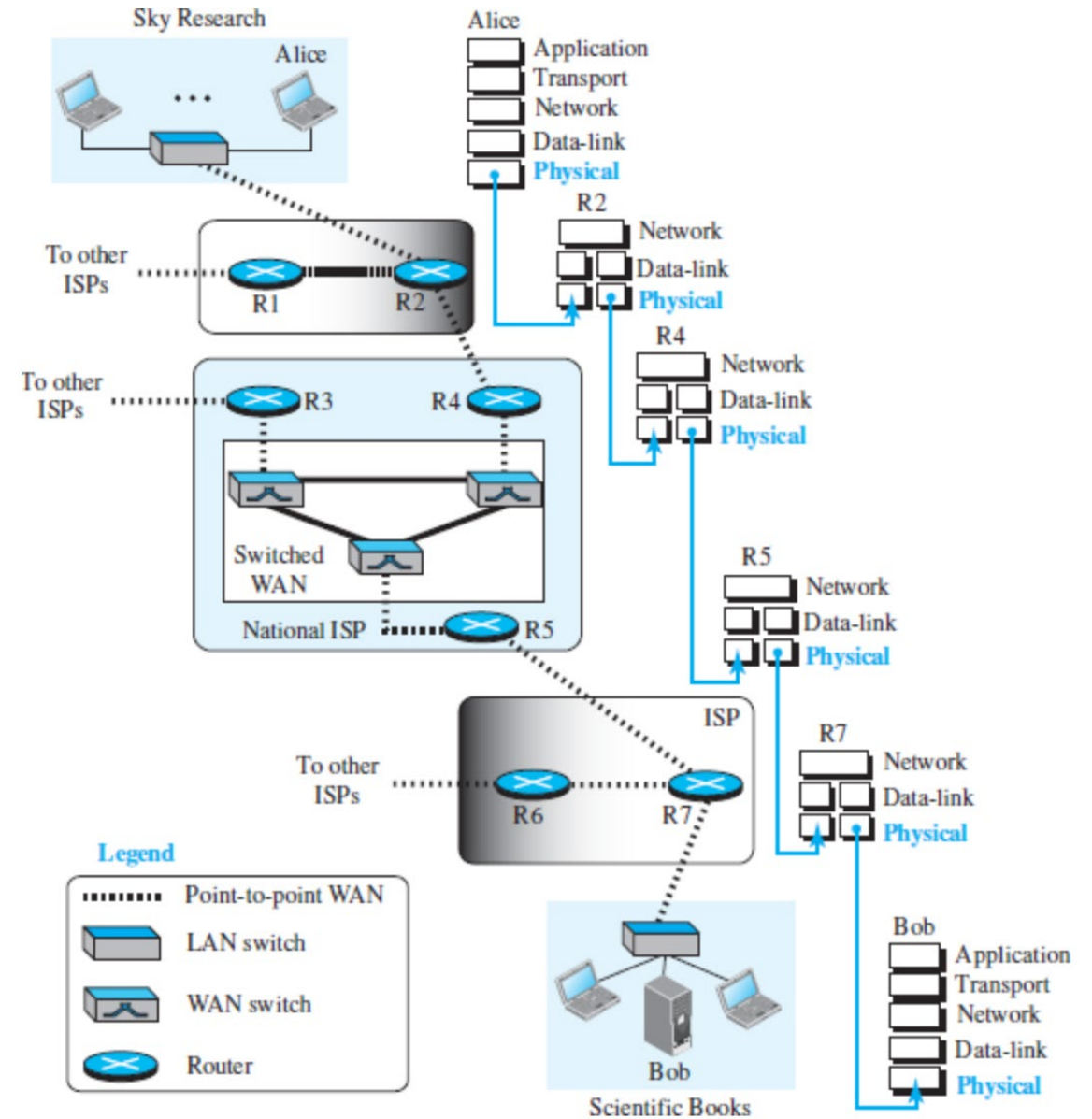
02. Week 2

Learning Outcomes

- By the end of this lecture, you will be able to:
 - Explain what are data and signal as well as their types.
 - Explain the characteristics of periodic analog signals.
 - Explain the characteristics of digital signals.

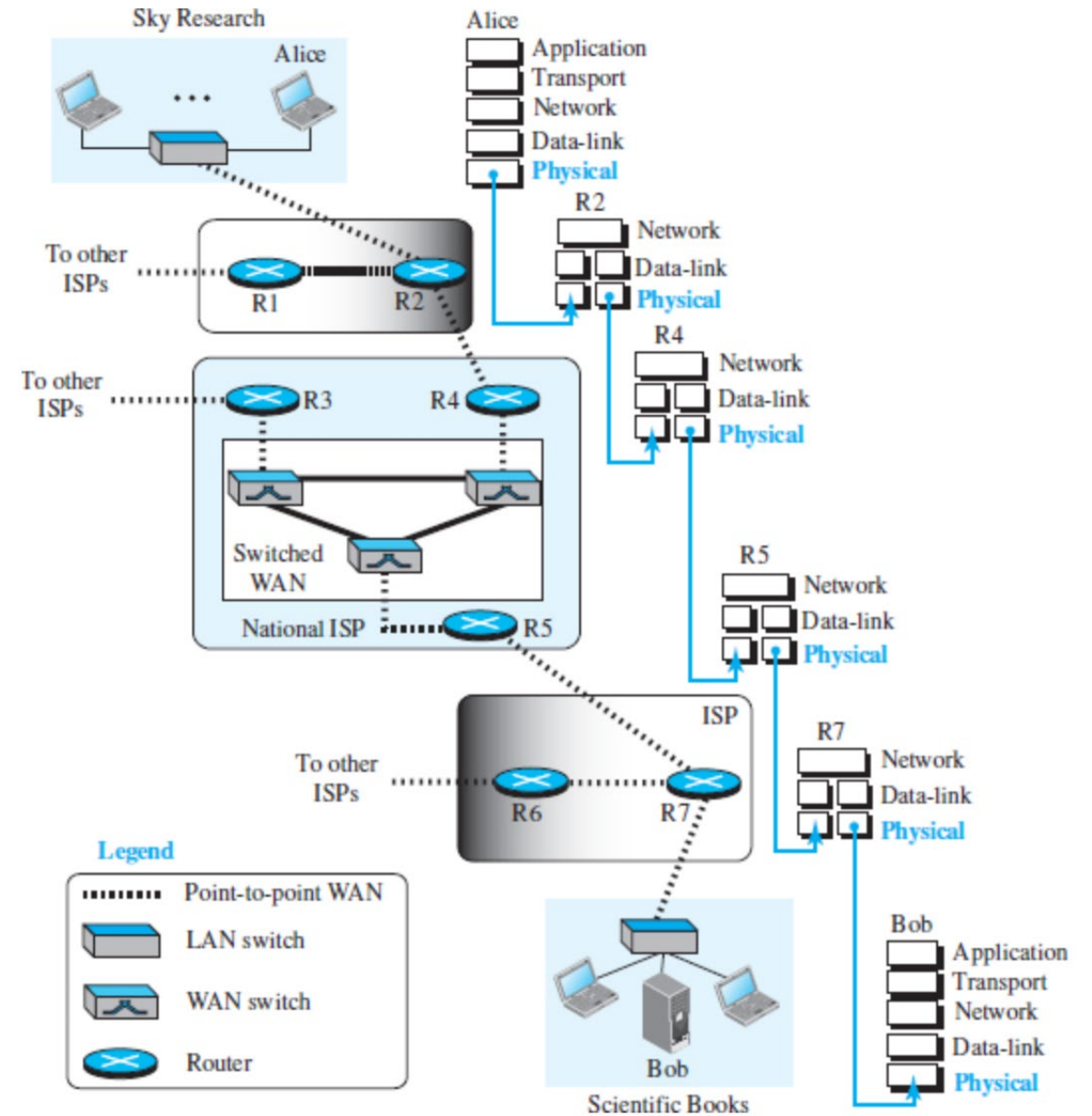
Introduction

- What are really exchanged between Alice and Bob?
- What goes through the network connecting Alice to Bob at the physical layer?



Introduction

- What are really exchanged between Alice and Bob?
 - **Data (information)**
- What goes through the network connecting Alice to Bob at the physical layer?
 - **Signals (e.g., electrical signals)**



Introduction

- **Physical layer**
 - Moving data in the form of **electromagnetic signals** across a **transmission medium**.
- **Data** must be changed to **signals** for **transmission**.
- **Communication** at application, transport, network, and data-link is **logical**.
- Communication at the physical layer is **physical**.

Analog and Digital Data

- **Analog data**

- Information that is **continuous** (takes on continuous values)
- **Real-life example:** **sound** (when someone speaks, an analog wave is created in the air)

- **Digital data**

- Information that has **discrete** states (takes on discrete values)
- Real-life example: **data are stored in computer memory** in the form of 1s and 0s

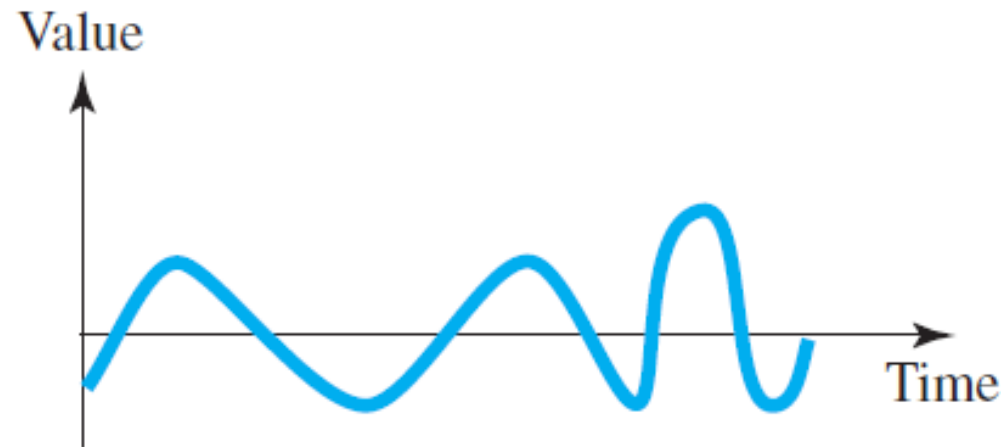


vs.

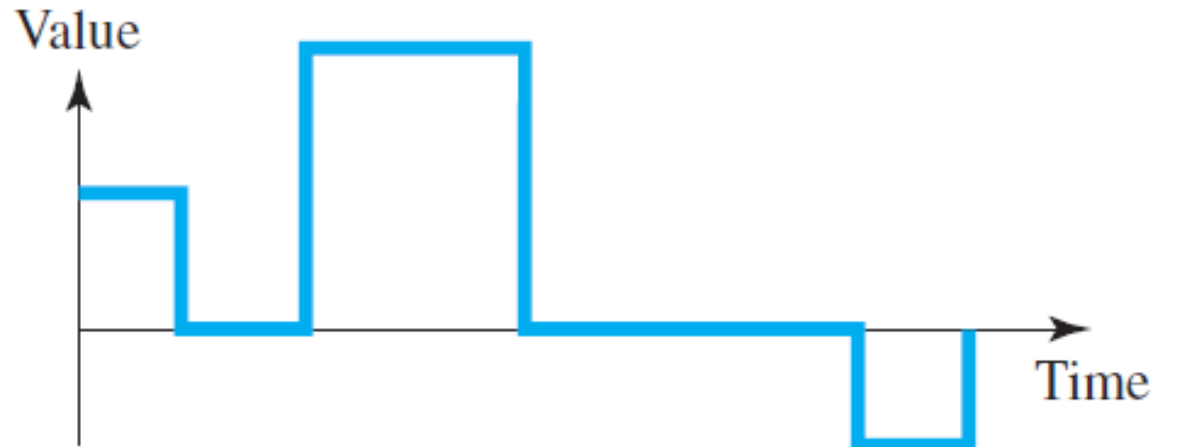


Analog and Digital Signals

- Analog signal
 - Has many levels of intensity over a period of time.
- Digital signal
 - Has a limited number of defined values (often 0 and 1).



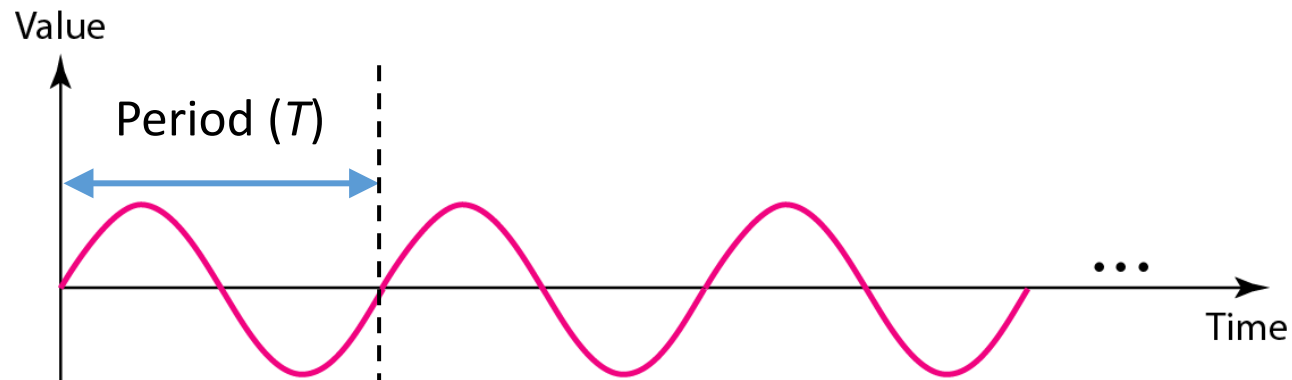
a. Analog signal



b. Digital signal

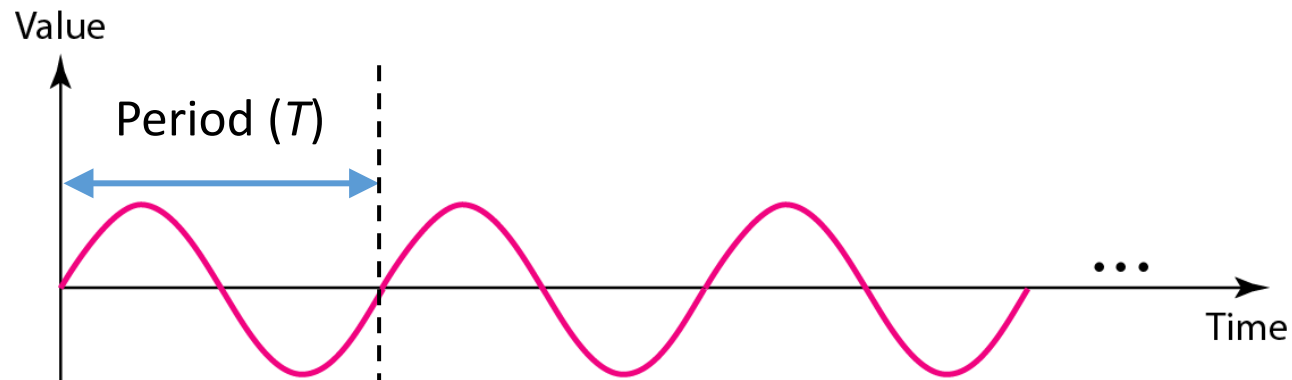
Periodic and Nonperiodic Signals

- Both analog and digital signals can take one of two forms:
 - **Periodic**
 - **Cycle**: the completion of one full pattern
 - **Period** (T): the amount of time, in seconds, a signal needs to complete one full pattern (i.e., one cycle)
 - A simple periodic analog signal, a **sine wave**, cannot be decomposed into simpler signals.
 - **Nonperiodic (Aperiodic)**



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 - **Nonperiodic (Aperiodic)**



In data communications, we commonly use **periodic analog signals** and **nonperiodic digital signals**.

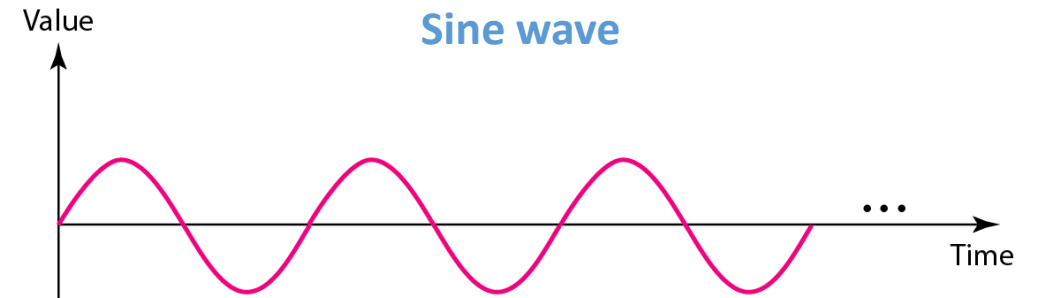
Periodic Analog Signals

Simple

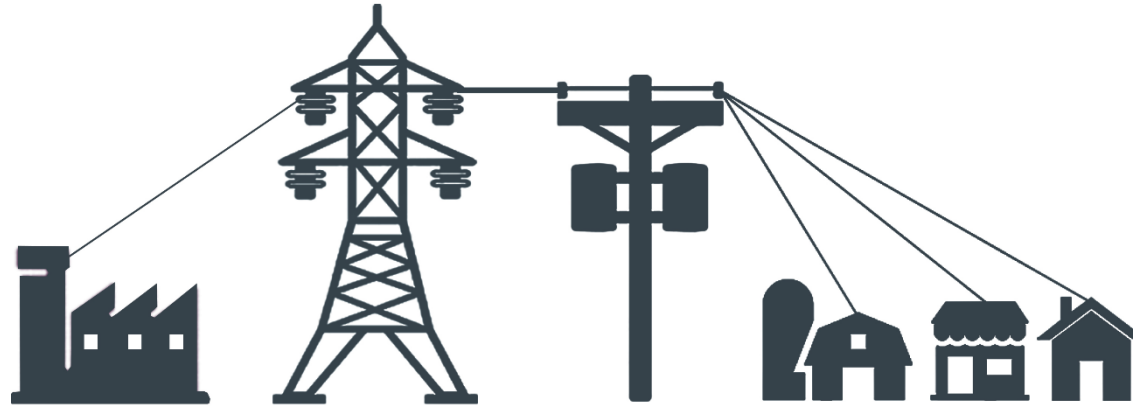
- Cannot be decomposed into simpler signals.

Composite

- Is composed of multiple sine waves.



Sine Wave – Real-life Applications



Power Distribution

The sine wave is carrying energy.

The sine wave is a signal of danger.



Burglar Alarm

Sine Wave

- The sine wave is the most fundamental form of a periodic analog signal.
- **Three** parameters that represent the sine wave:
 1. **Peak amplitude** (A): value of its highest intensity
 2. **Frequency** (f): # of completed cycles (periods) in 1s.
 3. **Phase** or **phase shift** (φ) : position of the waveform relative to time 0.
 - Phase is measured in degrees or radians (360° is 2π rad).

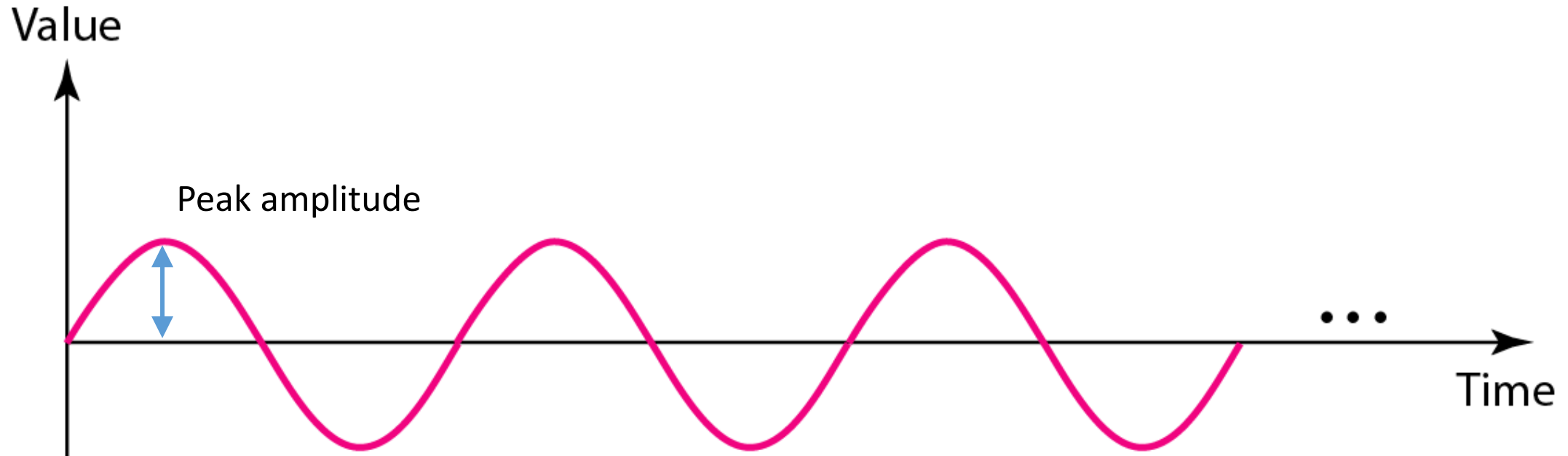
Example

- The electrical voltage in our homes in the Canada is periodic with a peak value about $120\sqrt{2} \cong 170$ V. Its frequency is 60 Hz.
- The voltage of a battery is constant (for example, 1.5 V).
 - Periodic with a frequency of 0.

Sine Wave – Peak Amplitude

1. Peak amplitude

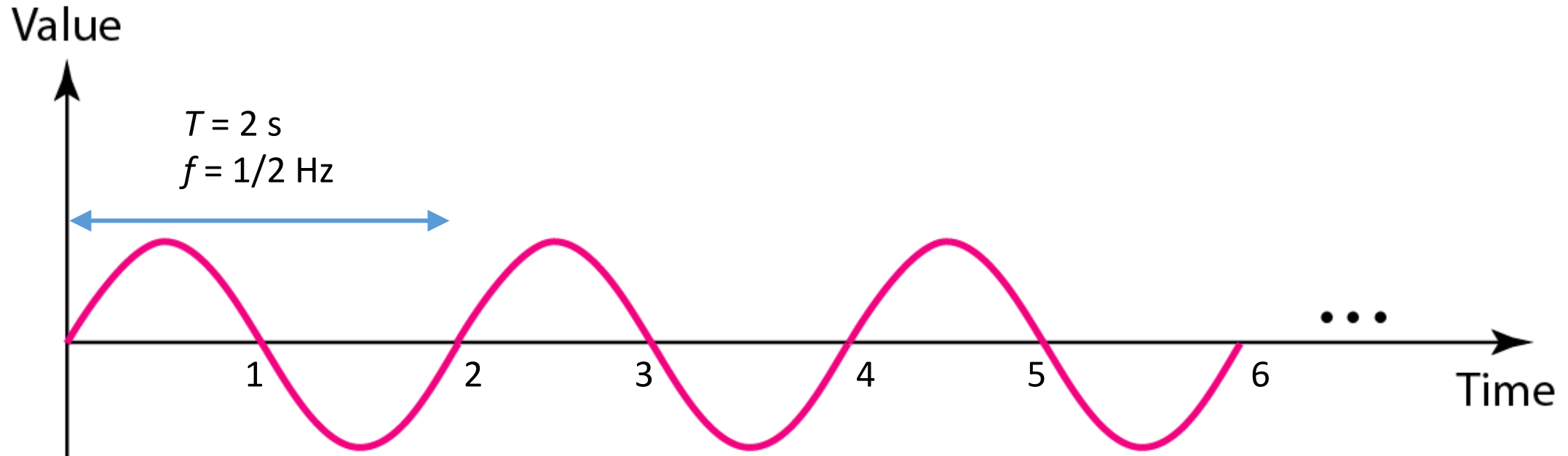
- The **absolute value of the signal's highest intensity**, proportional to the energy it carries.
- Measured in **volts**.



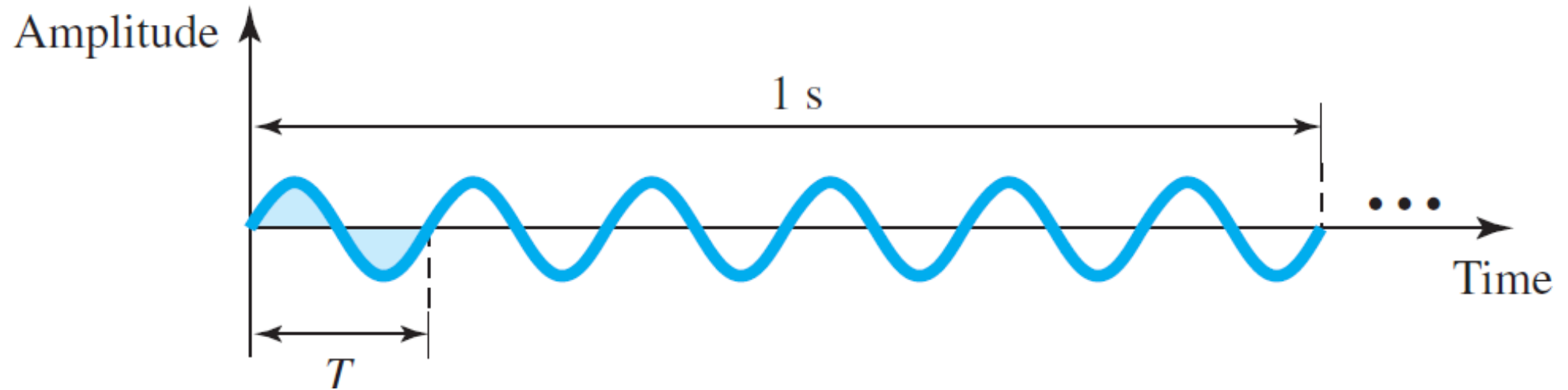
Sine Wave – Frequency

2. Frequency (f)

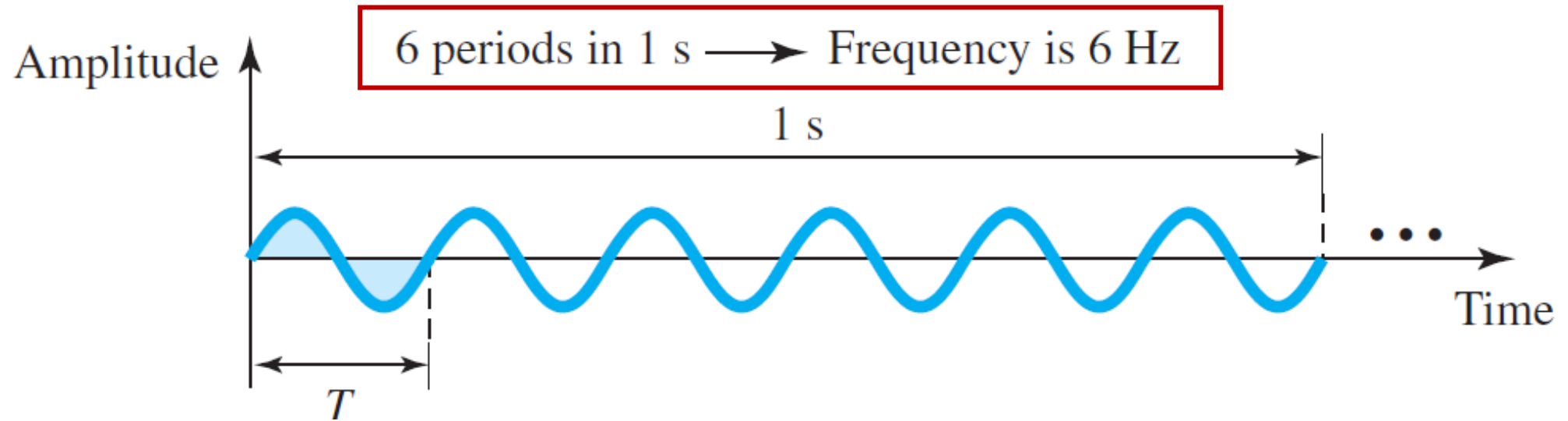
- The number of cycles in 1 second (the rate at which the signal repeats).
- Measured in **Hertz** (Hz) = **cycles per second**



Sine Wave – Frequency (Cont.)



Sine Wave – Frequency (Cont.)



Sine Wave – Frequency vs. Period

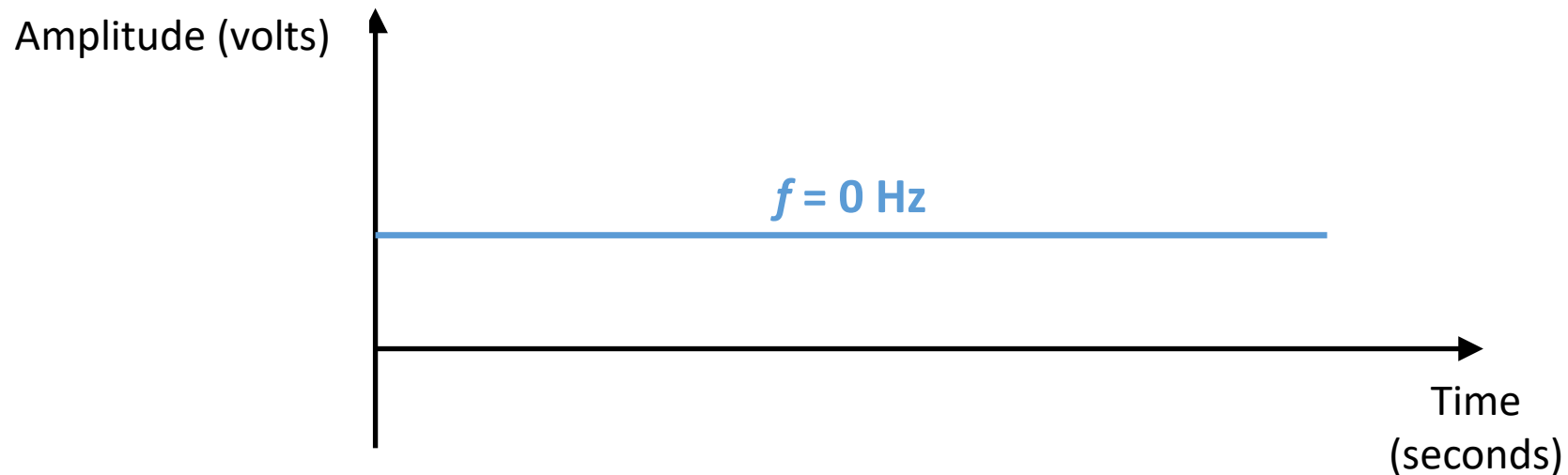
- **Period** and **Frequency** are just one characteristic described in two ways. Period is the inverse of frequency, and frequency is the inverse of period:

$$f = \frac{1}{T}$$

$$T = \frac{1}{f}$$

More about Frequency

- Frequency is the **rate of change** with respect to time.
 - Change in a **short span of time** means **high frequency**.
 - Change over a **long span of time** means **low frequency**.
 - If a signal **does not change at all**, its frequency is **zero**.
 - If a signal changes **instantaneously**, its frequency is **infinite** ($T = 0$ s).



Units of Period and Frequency

Period		Frequency	
Unit	Equivalent	Unit	Equivalent
Second (s)	1 s	Hertz (Hz)	1 Hz
Millisecond (ms)	10^{-3} s	Kilohertz (kHz)	10^3 Hz
Microsecond (μ s)	10^{-6} s	Megahertz (MHz)	10^6 Hz
Nanosecond (ns)	10^{-9} s	Gigahertz (GHz)	10^9 Hz
Picosecond (ps)	10^{-12} s	Terahertz (THz)	10^{12} Hz

Sine Wave – Frequency – Example 1

- The power we use at home has a frequency of 60 Hz. Find the period of this sine wave in milliseconds (ms)?

Sine Wave – Frequency – Example 1

- The power we use at home has a frequency of 60 Hz. Find the period of this sine wave in milliseconds (ms)?

- **Answer:**

$$T = 1/f = 1/(60 \text{ Hz}) = 0.0167 \text{ s} = \mathbf{16.7 \text{ ms}}$$

Sine Wave – Frequency – Example 2

- What is the frequency (in kHz) of a sine wave if the period is $200\ \mu\text{s}$?

Sine Wave – Frequency – Example 2

- What is the frequency (in kHz) of a sine wave if the period is 200 μs ?

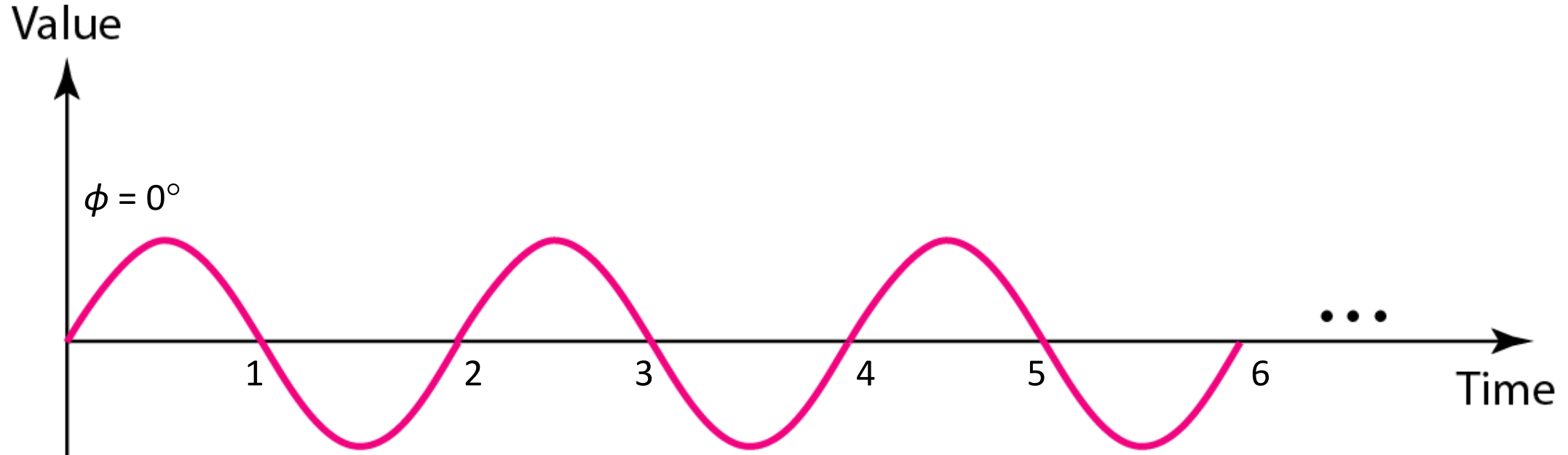
- **Answer:**

$$f = 1/T = 1/(200 \times 10^{-6} \text{ s}) = 1000000/200 \text{ Hz} = 5000 \text{ Hz} = \mathbf{5 \text{ kHz}}$$

Sine Wave – Frequency

3. Phase or Phase shift (ϕ)

- The position of the waveform relative to time 0 (indicates the status of the first cycle).
- Measured in **degrees** or **radians**.



Sine Wave – Mathematical Representation

- We can mathematically describe a sine wave as follows.

$$s(t) = A \sin(2\pi f t) = A \sin\left(\frac{2\pi}{T} t\right)$$

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The diagram illustrates the components of the sine wave equation $s(t) = A \sin(2\pi f t) = A \sin\left(\frac{2\pi}{T} t\right)$. Four blue arrows point from the variables in the equation to their corresponding physical meanings:

- An arrow from $s(t)$ points to **Instantaneous amplitude**.
- An arrow from A points to **Peak amplitude**.
- An arrow from f points to **Frequency**.
- An arrow from T points to **Period**.

Mathematical Representation – Example

- Find the peak amplitude, frequency, and period of the following sine waves.
 - $s(t) = 5\sin(20\pi t)$
 - $s(t) = \sin(10t)$

Mathematical Representation – Example

- Find the peak amplitude, frequency, and period of the following sine waves.

a. $s(t) = 5\sin(20\pi t)$

b. $s(t) = \sin(10t)$

- Answers:**

a. Peak amplitude: $A = 5 \text{ V}$

Frequency: $2\pi f = 20\pi \rightarrow f = 10 \text{ Hz}$

Period: $T = 1/f = 1/10 = 0.1 \text{ s}$

b. Peak amplitude: $A = 1 \text{ V}$

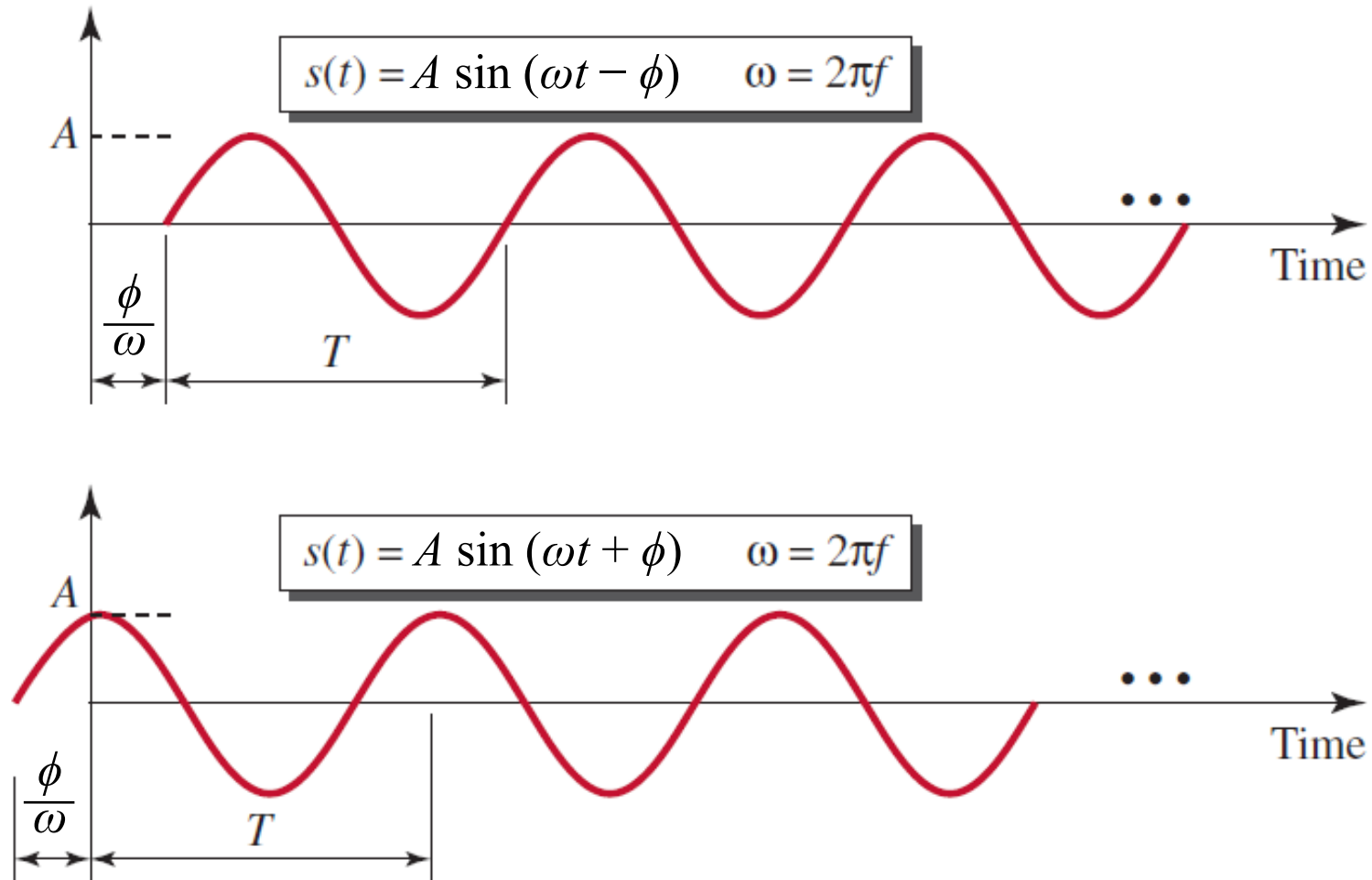
Frequency: $2\pi f = 10 \rightarrow f = 10/(2\pi) = 1.59 \text{ Hz}$

Period: $T = 1/f = 1/1.59 = 0.628 \text{ s}$

Shifting

- By replacing $2\pi f$ with ω in the sine wave mathematical representation, we have $s(t) = A\sin(\omega t)$. In this equation, the phase (i.e., phase shift) is zero. If we add or subtract a non-zero number ϕ to/from ωt , then our phase will be non-zero.

Horizontal Shifting (Phase Shift)

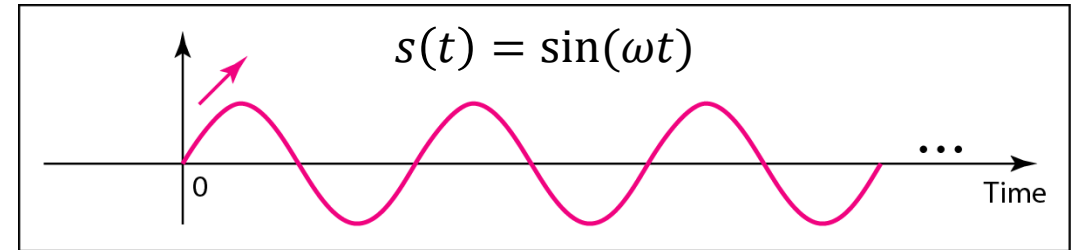


More about Phase

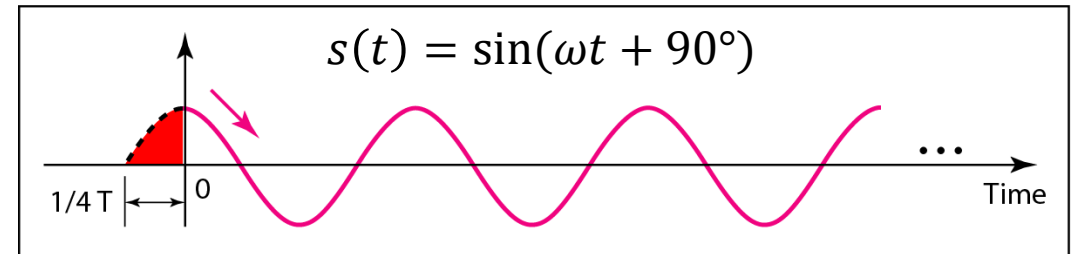
- $360^\circ = 2\pi \text{ rad}$
- $1^\circ = (2\pi/360) \text{ rad}$
- $1 \text{ rad} = (360/(2\pi))^\circ$
- A shift of a complete cycle is a phase shift of 360° .

Example 1

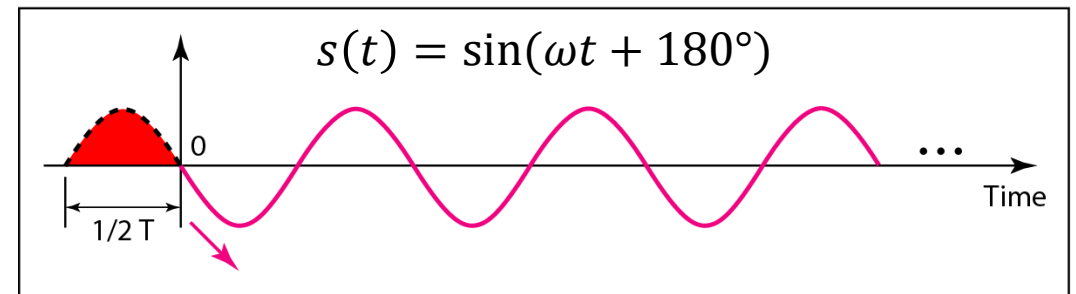
- Three sine waves with the same amplitude and frequency, but different phases.



a. 0 degrees



b. 90 degrees



c. 180 degrees

Example 2

- A sine wave is offset $1/9$ cycle with respect to time 0. What is its phase in degrees and radians?

Example 2

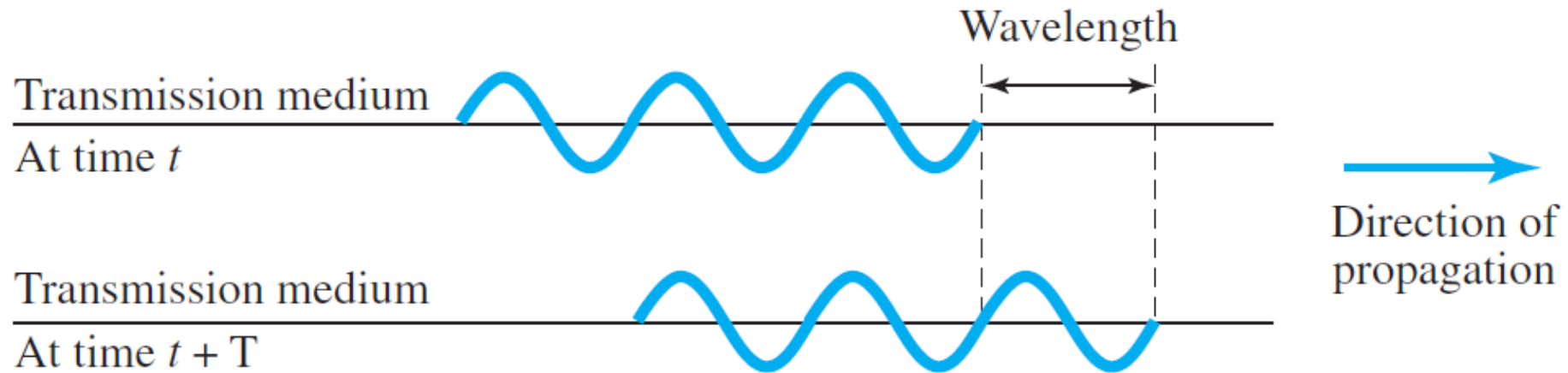
- A sine wave is offset $1/9$ cycle with respect to time 0. What is its phase in degrees and radians?

- **Answer:**

$$\begin{aligned}\phi &= \frac{1}{9} \times 360^\circ = 40^\circ \\ &= 40^\circ \times \frac{2\pi}{360^\circ} \text{ rad} = \frac{2\pi}{9} \text{ rad} = 0.698 \text{ rad}\end{aligned}$$

Wavelength

- Wavelength (λ)
 - The distance a simple signal can travel in one period.
 - (Distance that is travelled by a signal in 1 cycle.)
 - Usually used to describe the transmission of light in an optical fiber.
 - Usually measured in micrometres (μm).



Propagation Speed of a Signal

- Wavelength binds the period or the frequency of a simple sine wave to the **propagation speed** of the medium.
 - The propagation speed of electromagnetic signals depends on the medium and on the frequency of the signal.
 - For example, in a vacuum, light is propagated with a speed of $3 \times 10^8 \text{ m/s}$. That speed is lower in air and even lower in cable.

Wavelength (Cont.)

- **Frequency** vs **wavelength**
 - Frequency of a signal is **independent** of the **transmission medium**.
 - (so, what it really depends on?)
 - Wavelength relies on both **frequency** and **transmission medium**.

$$\lambda = \frac{c}{f} = c \times T$$

c is propagation speed.
 $c = 3 \times 10^8$ m/s (light)

Wavelength – Example

- What is the wavelength of red light if its frequency is 4×10^{14} Hz?
Assume the propagation speed is 3×10^8 m/s.

Wavelength – Example

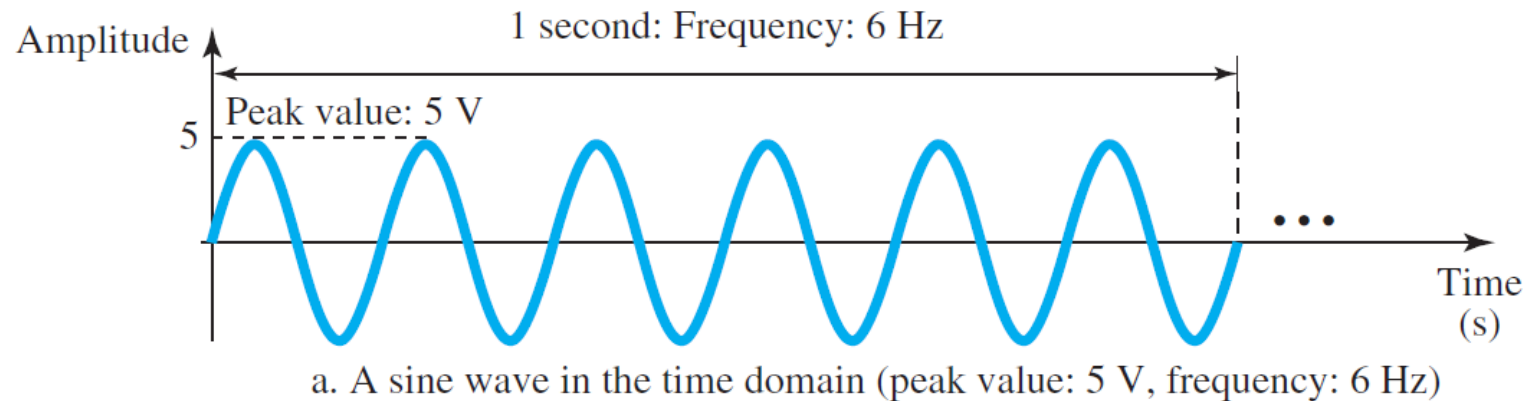
- What is the wavelength of red light if its frequency is 4×10^{14} Hz? Assume the propagation speed is 3×10^8 m/s.

- **Answer:**

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{4 \times 10^{14}} = 0.75 \times 10^{-6} = \mathbf{0.75 \mu m}$$

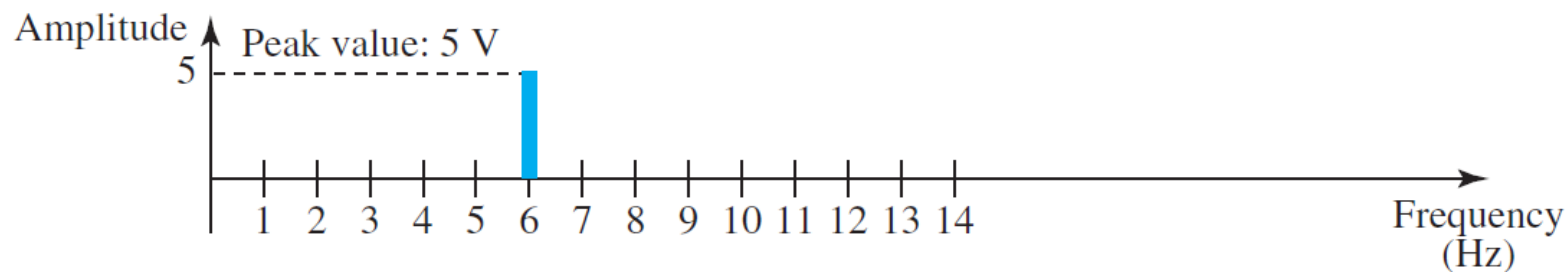
Time and Frequency Domains

- **Time-domain** plot
 - Changes in **signal amplitude** with respect to (w.r.t.) **time**.
 - Phase is not explicitly shown.



Time and Frequency Domains (Cont.)

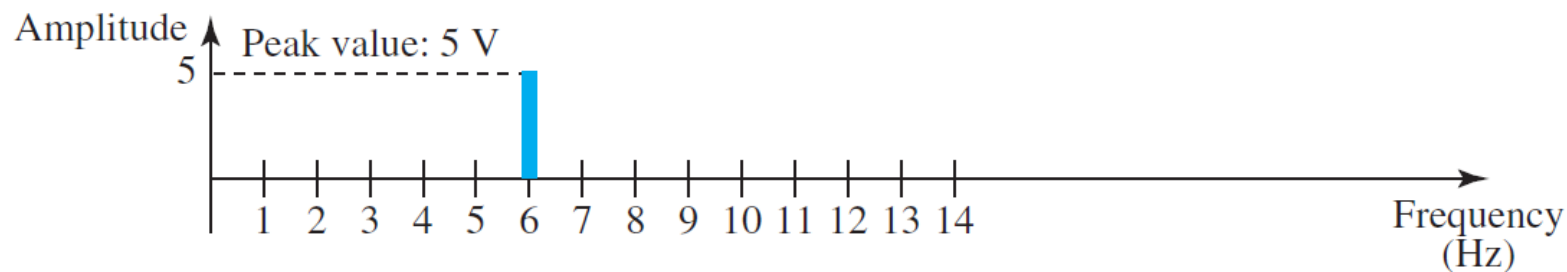
- **Frequency-domain** plot
 - Relationship between **amplitude** (**peak value**) and **frequency**.
 - **Advantage**: one can immediately see the values of the frequency and peak amplitude (a sine wave is represented by one spike).
 - More compact and helpful when dealing with **more than one sine wave**.



b. The same sine wave in the frequency domain (peak value: 5 V, frequency: 6 Hz)

Time and Frequency Domains (Cont.)

- **Frequency-domain** plot
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b. The same sine wave in the frequency domain (peak value: 5 V, frequency: 6 Hz)

A complete sine wave in the time domain can be represented by one single spike in the frequency domain.

Composite Signals

- Simple sine waves have many applications in daily life, such as **sending energy** from one place to another (**power distribution**).
- However, if we had only one single sine wave to convey a conversation over the phone, it would make no sense and carry no information. We would just hear a **buzz**. Thus:
 - We need to send a **composite signal** to communicate data.

Composite Signals

- **Composite signal**
 - A signal made of **many simple sine waves**.
 - A single-frequency sine wave is not useful in data communications!

Composite Signals

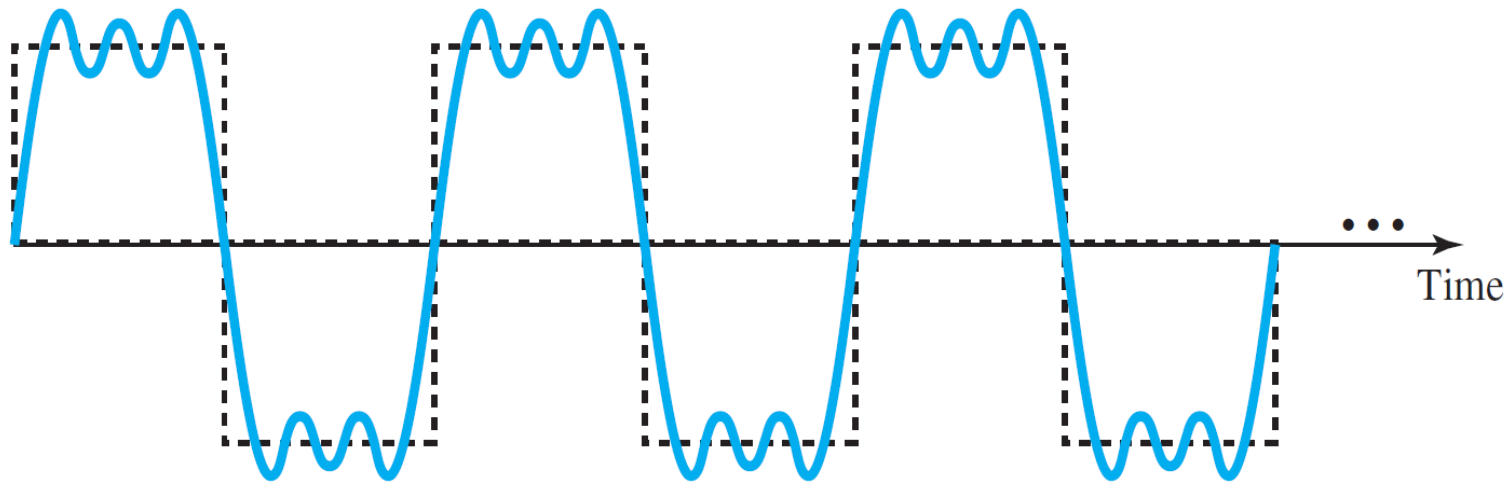
- **Composite signal**
 - A signal made of **many simple sine waves**.
 - A single-frequency sine wave is not useful in data communications!
- **Fourier analysis**
 - Any composite signal is a combination of simple sine waves with **different frequencies, peak amplitudes, and phases**.



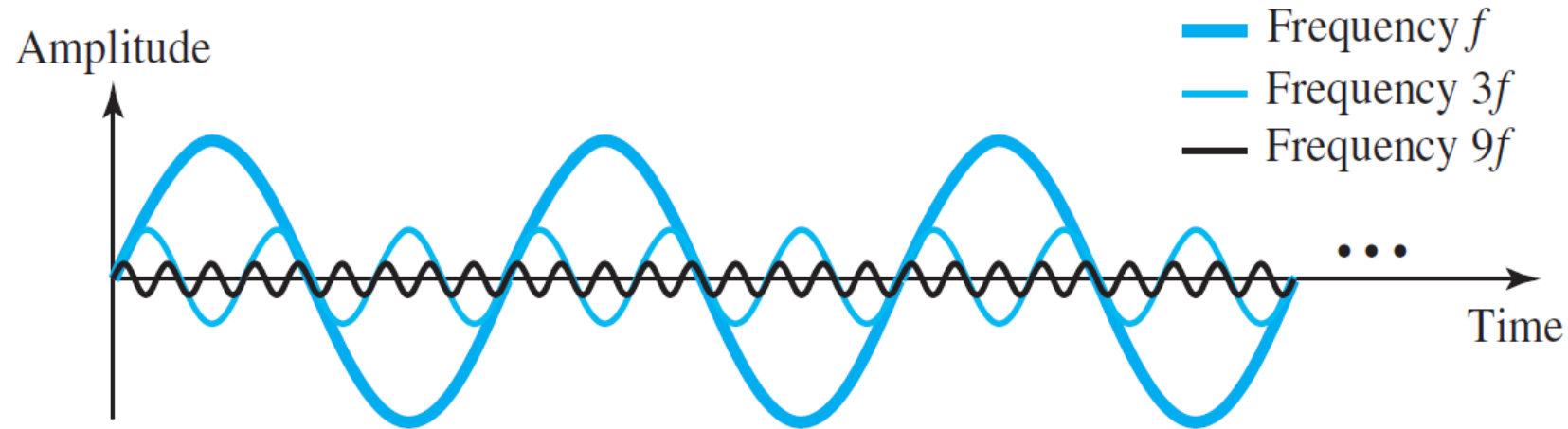
Jean-Baptiste Fourier

A Composite Periodic Signal

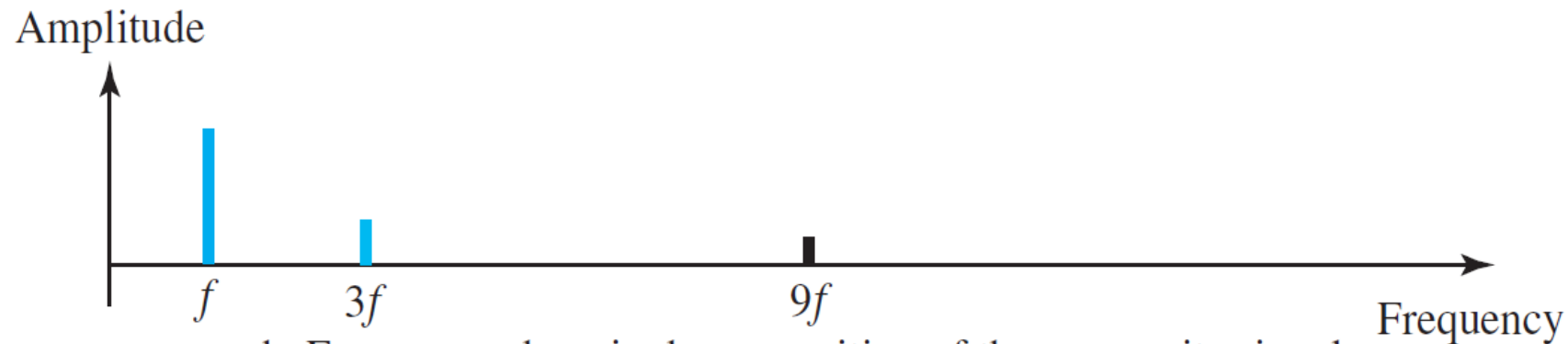
- **Composite periodic signal**
 - The decomposition gives a series of simple sine waves with **discrete frequencies** (frequencies with integer values). See next slide.



Decomposition of a Composite Periodic Signal

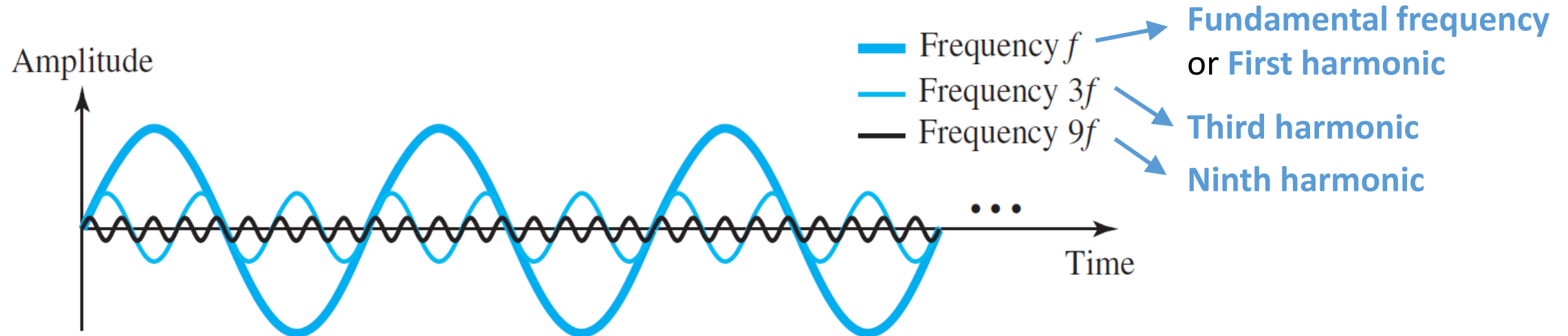


a. Time-domain decomposition of a composite signal

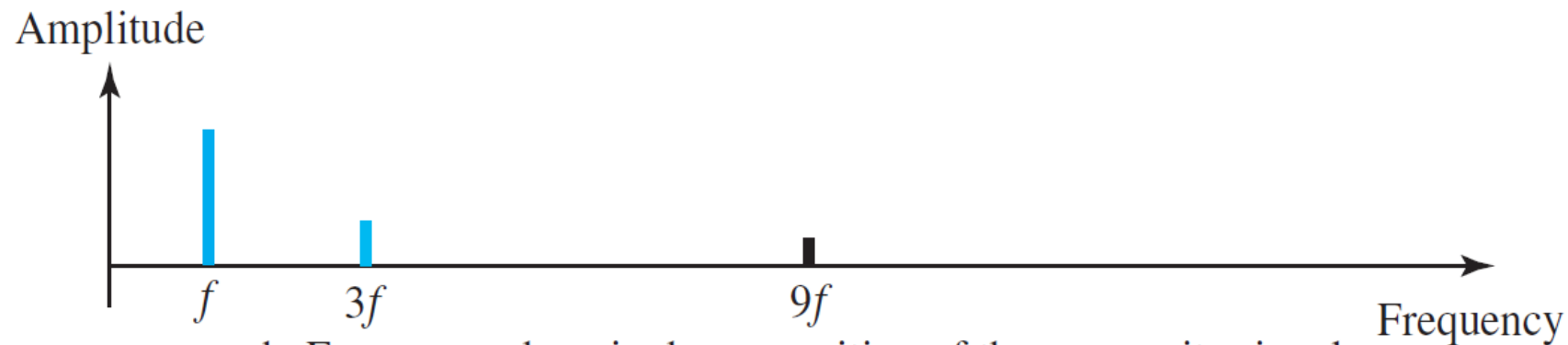


b. Frequency-domain decomposition of the composite signal

Decomposition of a Composite Periodic Signal



a. Time-domain decomposition of a composite signal

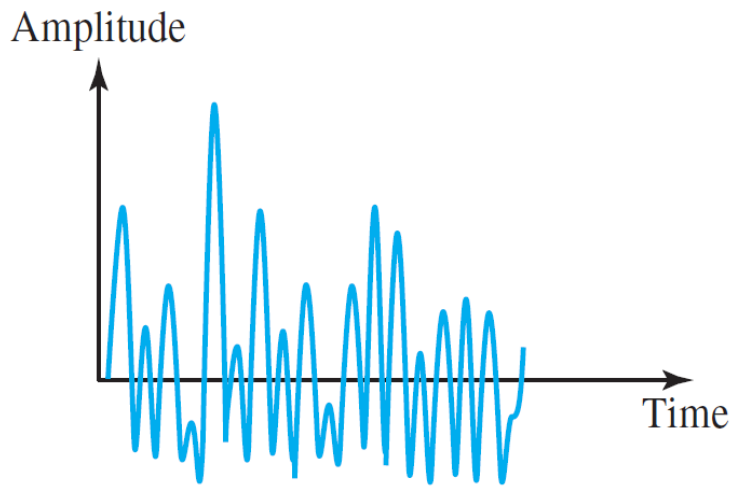


b. Frequency-domain decomposition of the composite signal

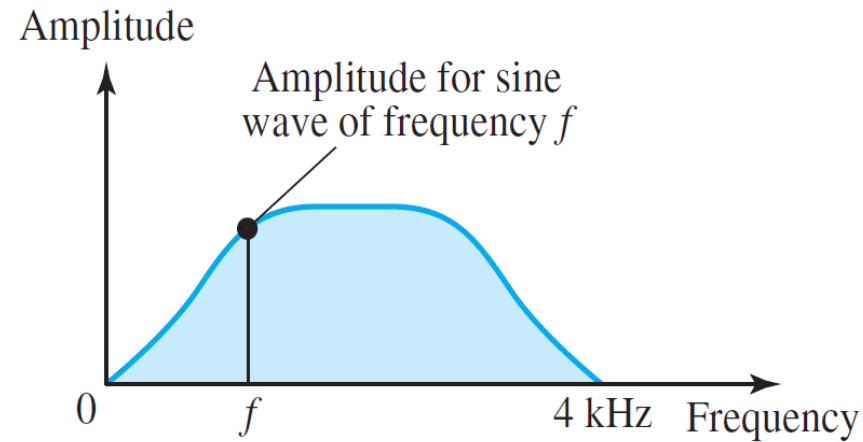
A Composite Nonperiodic Signal

- **Composite nonperiodic signal**

- The decomposition gives a combination of an **infinite number** of simple sine waves with continuous frequencies (frequencies with real values).
- Real-life examples:
 - Human voice (continuous range of frequencies between 0 and 4 kHz).
 - The signal propagated by an AM or FM radio station.



a. Time domain



b. Frequency domain

Bandwidth of a Composite Signal

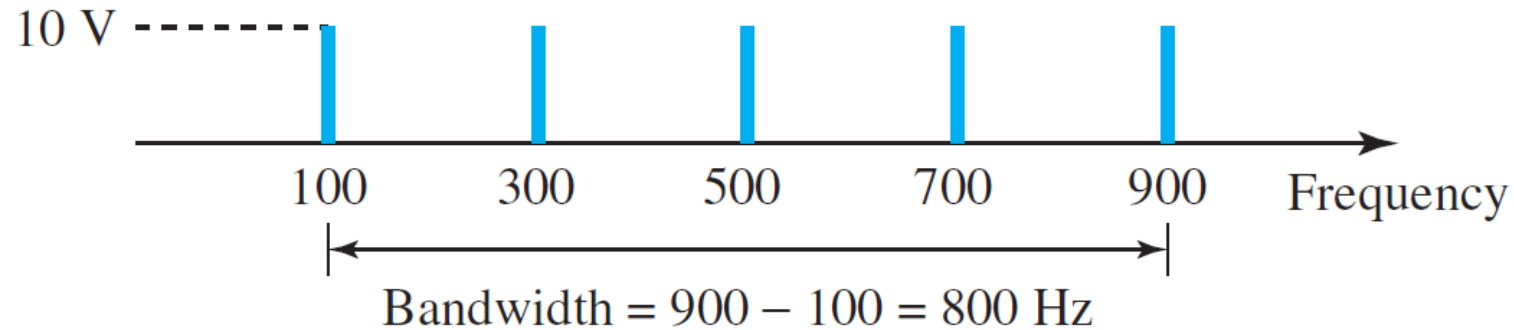
- Bandwidth (B): the **difference** between the **highest** and the **lowest frequencies** contained in a composite signal.

$$B = f_h - f_l$$

Bandwidth of a Composite Signal

- Bandwidth (B): the **difference** between the **highest** and the **lowest frequencies** contained in a composite signal.

$$B = f_h - f_l$$



Digital Signals

- Most digital signals are **nonperiodic**.
 - Frequency and period are not suitable characteristics.
- **Bit rate** is used to describe digital signals.
 - Defined as **the number of bits sent per second**, expressed in **bits per second (bps)**.
- **Bit length** (a similar concept to wavelength)
 - Defined as the **distance one bit occupies** on the transmission medium.
- **Bit duration**: $1/(\text{bit rate})$
 - E.g., $1/1 \text{ Mbps} = 1 \mu\text{s}$

$$\text{Bit length} = \text{propagation speed} \times \text{bit duration}$$

Example

- What is the bit length of a signal that has a bit rate of 1 Mbps and is travelling at 2×10^8 m/s on a transmission medium.

$$\text{Bit length} = \text{propagation speed} \times \text{bit duration}$$

Example

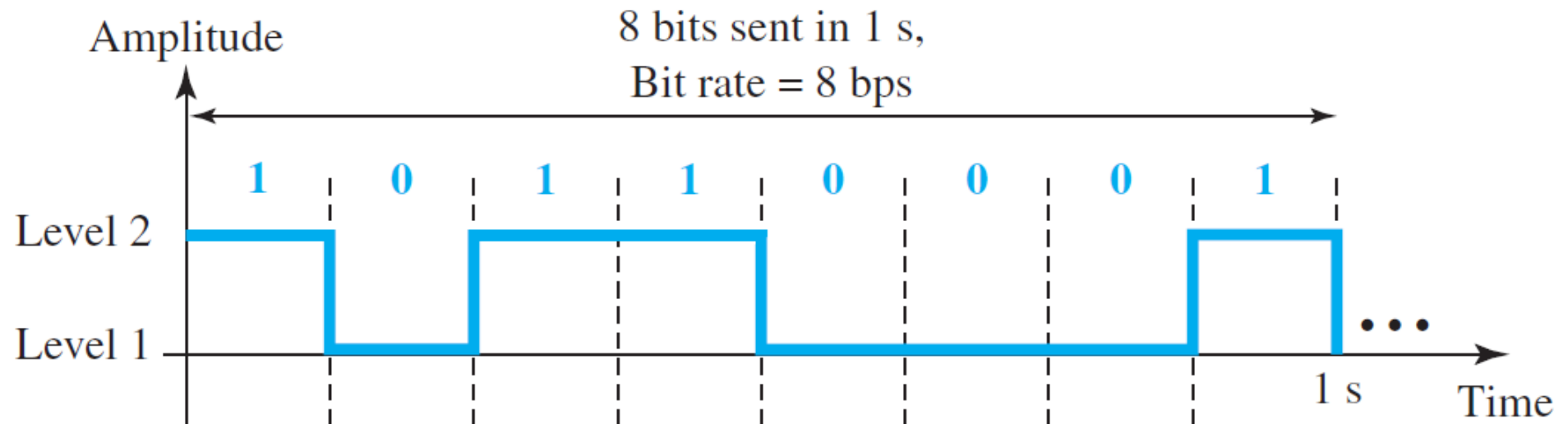
- What is the bit length of a signal that has a bit rate of 1 Mbps and is travelling at 2×10^8 m/s on a transmission medium.
- **Answer:**
 - Bit duration = $1/(1 \text{ Mbps}) = 1 \mu\text{s}$
 - Bit length = $(2 \times 10^8 \text{ m/s}) \times 1 \mu\text{s} = 200 \text{ m}$
 - This means a bit occupies 200 meters on this transmission medium.

$$\text{Bit length} = \text{propagation speed} \times \text{bit duration}$$

Digital Signals – Level

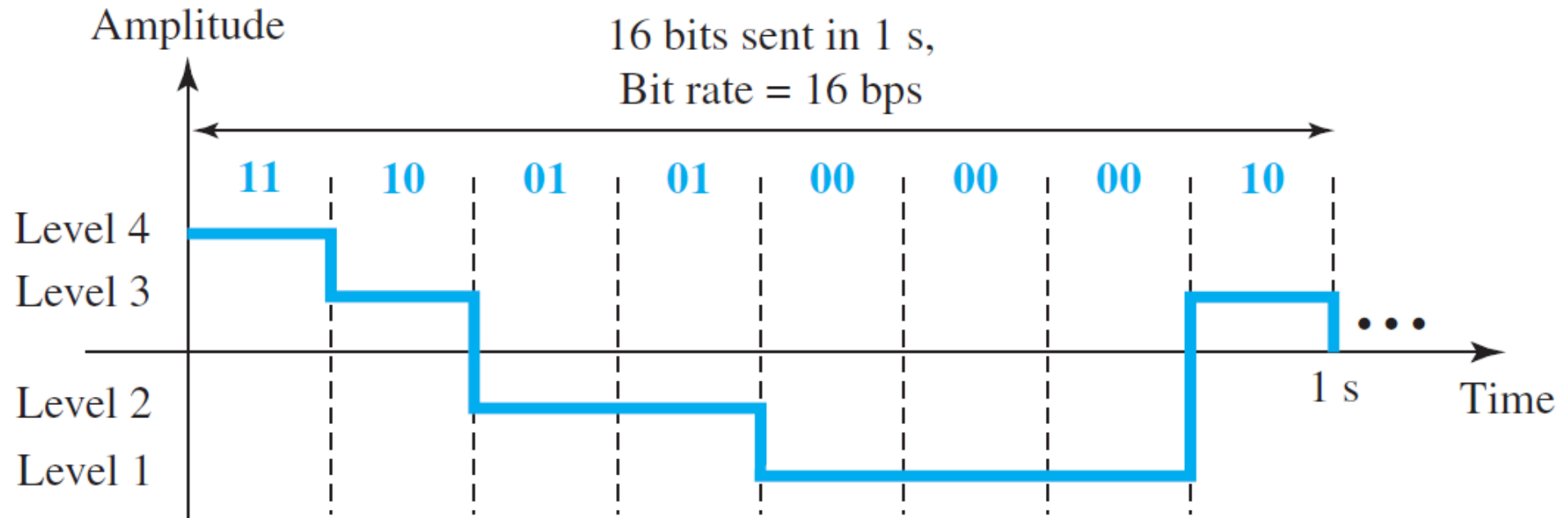
- “**Level**” refers to a specific state or value that a digital signal can have at a given point in time.
- Digital signals are characterized by having **discrete** levels or states, each of which represents a distinct value or symbol. These levels are typically associated with **voltage** or **current** levels in electronic circuits.
 - In **binary** digital systems, there are usually two signal levels: a “low” level (often represented as 0) and a “high” level (often represented as 1).
 - In more advanced digital systems, you can have even more signal levels, such as octal (eight levels) or hexadecimal (sixteen levels).

A Digital Signal with Two Levels



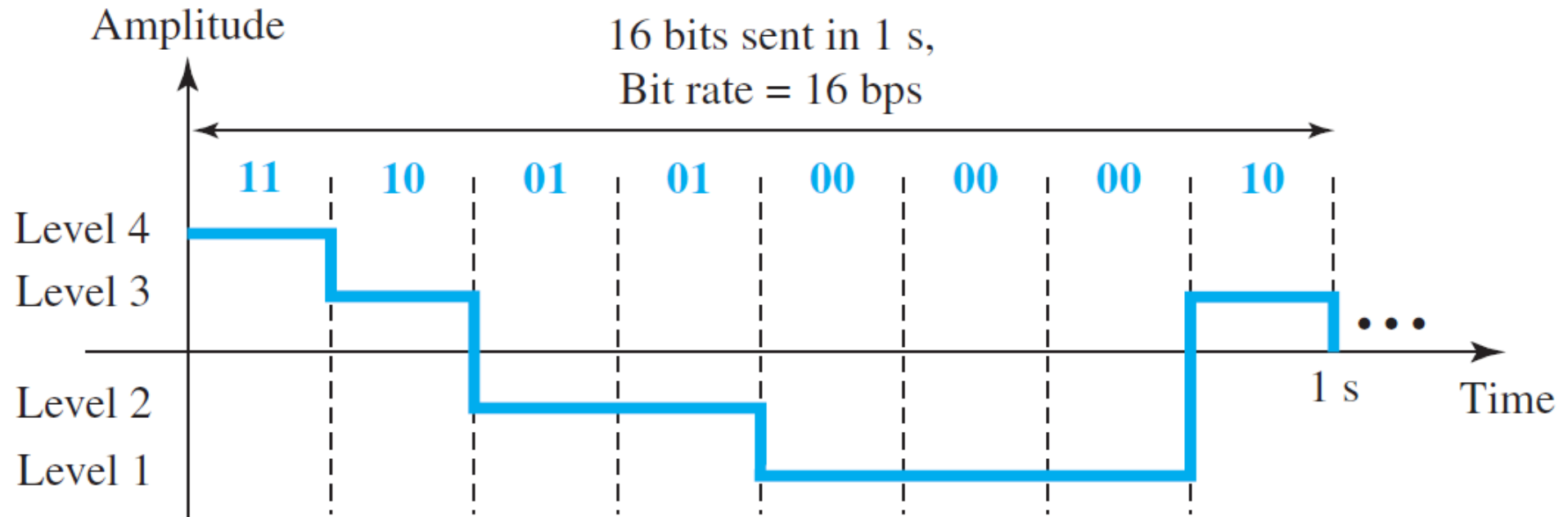
a. A digital signal with two levels

A Digital Signal with Four Levels



b. A digital signal with four levels

A Digital Signal with Four Levels



b. A digital signal with four levels

To encode 4 levels, $\log_2 4 = 2$ bits are required.

Digital Signals – Example (Signal Levels)

- A digital signal has 11 levels. How many bits are needed?

Digital Signals – Example (Signal Levels)

- A digital signal has 11 levels. How many bits are needed?
- **Answer:**
 - $\log_2 11 = 3.46 \text{ bits}$
 - However, this answer is not realistic.
 - The number of bits needed has to be an integer and usually as a power of 2. For this example, **4 bits** should be used in this case.

Digital Signals

- For a signal with L levels, **number of bits needed** = $\lceil \log_2 L \rceil$
- **Ceiling** function ($\lceil x \rceil$): rounds the number up to the nearest integer greater than or equal to the original value.
 - E.g., ceiling of π : $\lceil \pi \rceil = \lceil 3.1416 \rceil = 4$
- **Floor** function ($\lfloor x \rfloor$): rounds the number down to the nearest integer less than or equal to the original value.
 - E.g., floor of π : $\lfloor \pi \rfloor = \lfloor 3.1416 \rfloor = 3$

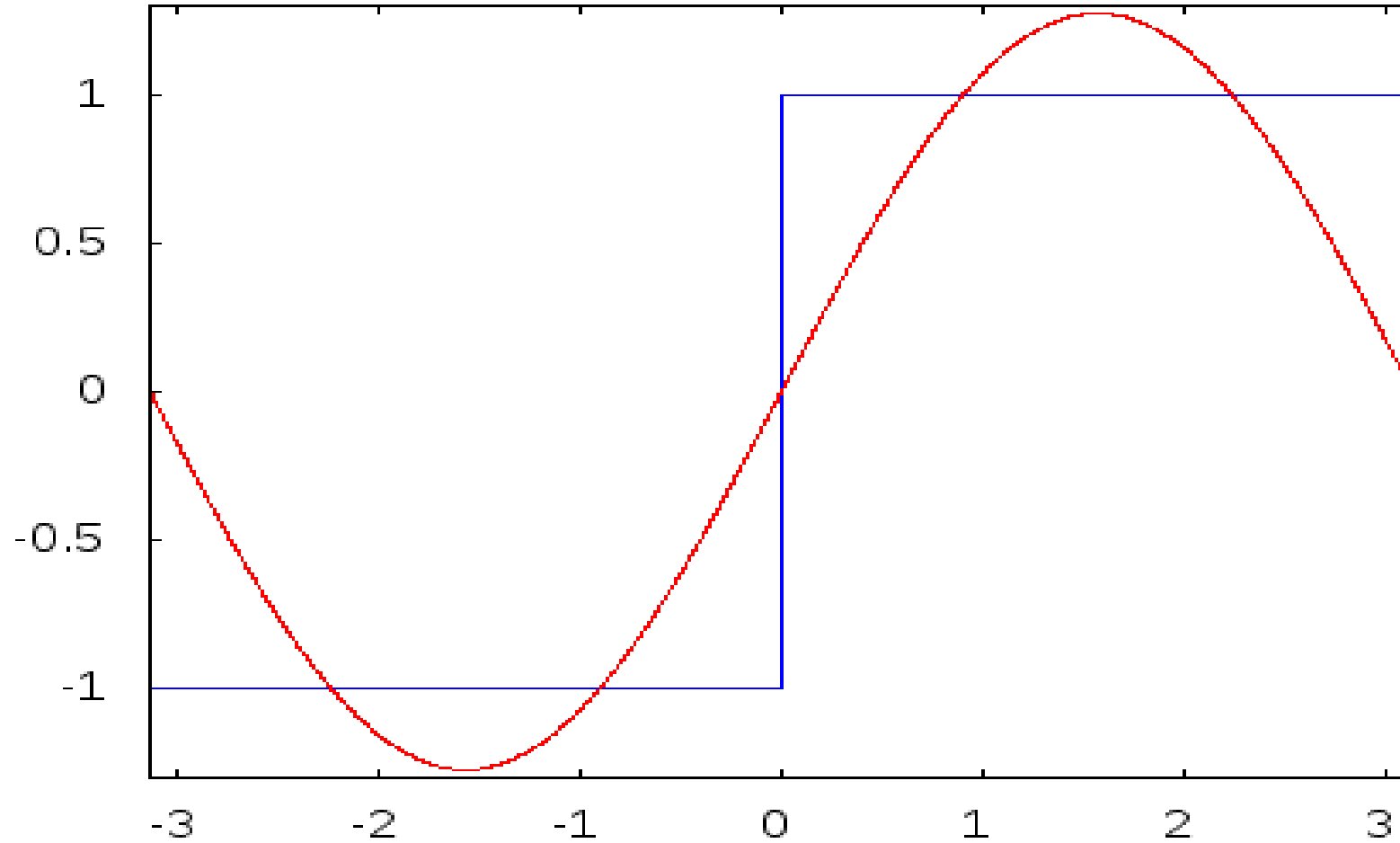
Digital Signals – Example (Bit Rate)

- Assume we need to download text documents at the rate of 100 pages per second. What is the required bit rate of the channel? A page is an average of 24 lines with 80 characters in each line. If we assume that one character requires 8 bits, the bit rate is:
- **Answer:**
 $100 \times 24 \times 80 \times 8 = 1536000 \text{ bps} = \mathbf{1.536 \text{ Mbps}}$

Digital Signal as a Composite Analog Signal

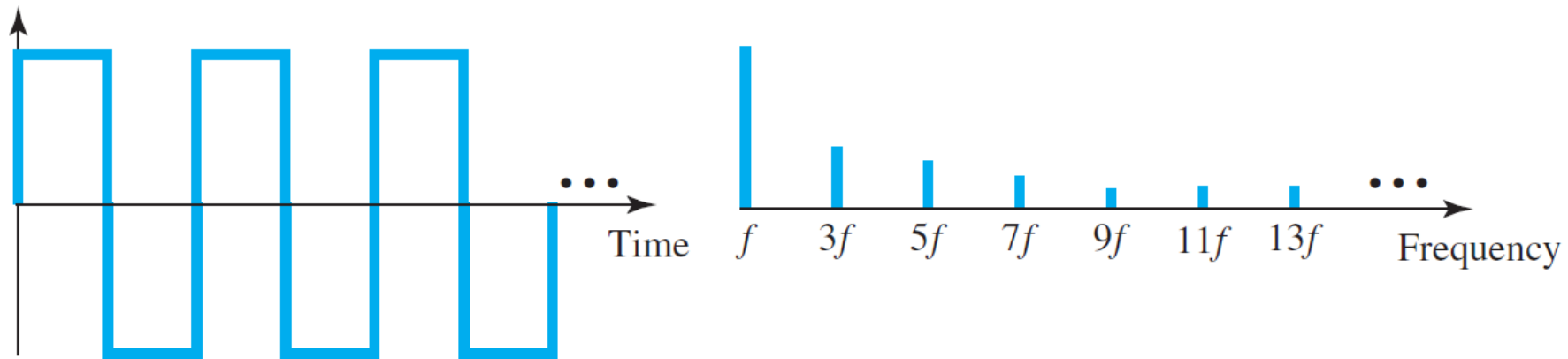
- A periodic or nonperiodic digital signal is a **composite analog signal** with frequencies between zero and infinity (**infinite bandwidth**).
- **Fourier analysis** can be used to decompose a digital signal.

Example



Periodic Digital Signal as a Composite Analog Signal

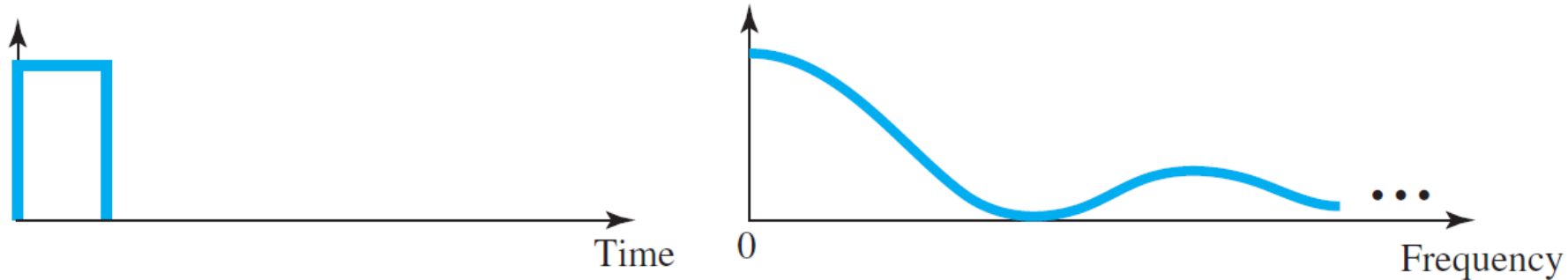
- **Periodic** digital signal (rare in data communications)
 - In **frequency** domain representation of this signal:
 - **Infinite bandwidth** and **discrete** frequencies



a. Time and frequency domains of periodic digital signal

Nonperiodic Digital Signal as a Composite Analog Signal

- **Nonperiodic** digital signal
 - In **frequency** domain representation of this signal:
 - **Infinite bandwidth** and **continuous** frequencies



b. Time and frequency domains of nonperiodic digital signal

Summary

- Transformation of data to electric signals for transmission.
- Types of data and signals as well as their characteristics.
- Analog signals and their characteristics.
- Digital signals and their characteristics.

References

[1] Behrouz A. Forouzan, Data Communications & Networking with TCP/IP Protocol Suite, 6th Ed, 2022, McGraw-Hill companies.

Reading

- Chapter 2 of the textbook, section 2.1
- Chapter 2 of the textbook, section 2.8 (Practice Test)